Chapter 13 — Halogens

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Volatility, bond strength and oxidizing power

Q.1 Explain the trends in volatility, bond strength and oxidizing power of the halogens.

Halogens are Group 17 diatomic non-metals. Chlorine is a greenish-yellow gas, bromine a red-brown liquid and iodine a greyish-black solid that gives violet vapour. Volatility decreases from Cl_2 to I_2 because larger molecules have stronger London dispersion forces.

Bond strength generally decreases from Cl_2 to I_2 as bond length increases. Fluorine is exceptional: the F–F bond is weak because lone-pair repulsion is strong in the very small molecule. Oxidizing power decreases down the group: $F_2 > Cl_2 > Br_2 > I_2$. A more reactive halogen displaces a less reactive halide from solution.

Textbook check (p. 285): the sentence “Chlorine being a liquid is very volatile” is a printing error; chlorine is a gas at room temperature.

Important Conceptual Short Questions — Topic 1

1. Why does iodine have a higher boiling point than bromine?

Iodine molecules are larger and more polarizable, so London dispersion forces are stronger and more energy is needed to separate them.

2. Why is fluorine a stronger oxidizing agent than chlorine?

Fluorine has a greater tendency to gain electrons, supported by its high electronegativity and very negative electrode potential, despite its weak F–F bond.

3. How can chlorine displace bromine from potassium bromide solution?

Chlorine is a stronger oxidizing agent than bromine, so it oxidizes Br^- to Br_2: $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$.

4. Why does F_2 have an unexpectedly weak bond?

The fluorine atoms are very small, so lone pairs on adjacent atoms repel strongly, weakening the F–F bond.

Topic 2 — Reactions with hydrogen and hydrogen halides

Q.2 Describe reactions of halogens with hydrogen and the thermal stability and reducing behaviour of hydrogen halides.

Halogens react with hydrogen to form hydrogen halides: $H_2 + X_2 \rightarrow 2HX$. Fluorine reacts explosively even in the dark, chlorine reacts in sunlight, bromine reacts on heating, and iodine reacts slowly and reversibly with a catalyst.

Thermal stability decreases from HF to HI because the H–X bond becomes longer and weaker. Reducing power of halide ions increases down the group: $I^- > Br^- >$

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Cl^- > F^-, because larger ions lose electrons more easily.

Important Conceptual Short Questions — Topic 2

1. Why does HI decompose more easily than HCl?

The H–I bond is longer and weaker than the H–Cl bond, so less energy is needed to break it.

2. Why is iodide ion a stronger reducing agent than chloride ion?

Iodide is larger and its outer electron is less strongly held, so it is more easily oxidized to iodine.

3. How does bond strength explain the trend in thermal stability of HX?

Stronger H–X bonds require more heat to break, so HF is most stable and HI least stable.

4. Why does the reaction of iodine with hydrogen remain reversible?

The H–I bond is weak and the reaction is not strongly favoured, so HI can decompose back into H_2 and I_2 under the same conditions.

Topic 3 — Tests for halide ions and reactions with sulfuric acid/alkali

Q.3 Describe tests for halide ions and reactions of halides with concentrated sulfuric acid and chlorine with alkali.

Silver nitrate solution identifies halide ions: AgCl is white and soluble in dilute ammonia, AgBr is cream and soluble in concentrated ammonia, and AgI is yellow and insoluble in ammonia. AgF is soluble, so no precipitate forms with fluoride.

With concentrated H_2SO_4, chlorides give HCl gas; bromides give HBr, Br_2 and SO_2; iodides give HI, I_2 and further reduction products such as H_2S. Chlorine disproportionates in cold dilute NaOH to NaCl and NaClO, and in hot concentrated NaOH to NaCl and $NaClO_3$.

Textbook check (pp. 291–292): the printed equation $2HBr + H_2SO_4 \rightarrow Br_2 + 2SO_2 + 2H_2O$ is not balanced; the balanced equation is $2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$. The hot-alkali equation should be $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$.

Important Conceptual Short Questions — Topic 3

1. Why does AgI not dissolve in ammonia?

AgI is too insoluble for ammonia to form enough soluble silver-ammine complex to dissolve it.

2. How can silver nitrate and ammonia distinguish Cl^-, Br^- and I^-?

AgCl is white and dissolves in dilute NH_3; AgBr is cream and dissolves in concentrated NH_3; AgI is yellow and does not dissolve.

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3. Why does concentrated sulfuric acid oxidize bromide and iodide but not chloride?

Br^- and I^- are stronger reducing agents than Cl^-, so they can reduce H_2SO_4 while being oxidized to Br_2 or I_2.

4. Why does chlorine give different products with cold and hot NaOH?

Temperature changes the stability of the products. Cold alkali favours hypochlorite, ClO^-, while hot alkali favours chlorate, ClO_3^-.

Topic 4 — Chlorine in water purification

Q.4 Explain the use of chlorine in water purification and factors affecting disinfection.

Chlorine is added to water to kill bacteria and other pathogens. It hydrolyses: $Cl_2 + H_2O \rightleftharpoons HCl + HOCl$. Hypochlorous acid, HOCl, partly dissociates to H^+ and OCl^-. HOCl is the more effective disinfectant because it penetrates microbial cells more readily.

Disinfection depends on pH, chlorine dose, contact time, temperature and organic matter. Lower pH favours HOCl, while higher pH favours the less effective OCl^-. A residual chlorine level helps protect water during distribution.

Important Conceptual Short Questions — Topic 4

1. Why is HOCl more effective as a disinfectant than ClO^-?

HOCl is neutral and can penetrate microbial cell walls more easily, where it oxidizes essential cellular components.

2. How does pH affect chlorine disinfection?

At lower pH more chlorine is present as HOCl, which is a stronger disinfectant. At higher pH more is present as OCl^-, which is less effective.

3. Why is residual chlorine important in water supplies?

It provides continued protection against microbial contamination as water travels through pipes to consumers.

4. How can organic matter reduce the efficiency of chlorination?

Organic matter consumes chlorine and can shield microorganisms, so less active chlorine remains available for disinfection.

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